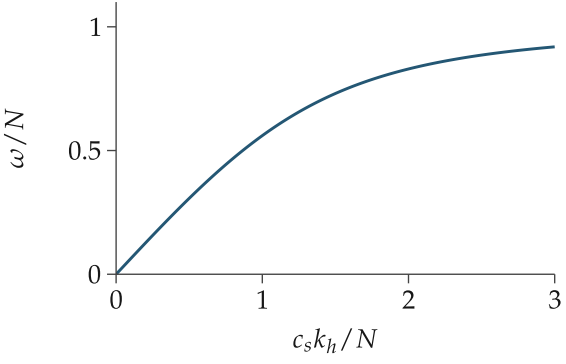
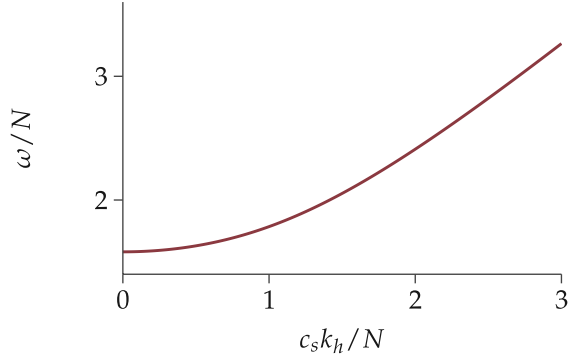
$$\omega^4 - \left[c_s^2(k_h^2 + m^2) + \omega_a^2\right]\omega^2 + c_s^2 N^2 k_h^2 = 0$$

a) Gravity branch



b) Acoustic branch



$$\omega_\pm^2 = \frac{S \pm \sqrt{S^2 - 4c_s^2 N^2 k_h^2}}{2}, \qquad S = c_s^2(k_h^2 + m^2) + \omega_a^2$$